12.6 Surface Areas and Volumes of Spheres

Name _____ Period ____ Date _____

Geometry 12.6 (30-minute plan) · Student sheet

I can…

- find the surface area of a sphere.
- find the volume of a sphere.
- find the surface area and volume of a hemisphere.

Vocabulary · fill in the missing word

Sphere — the set of all points in space that are the same _____ from a given point, the center

Great circle — the _____ of a sphere and a plane that contains the center of the sphere

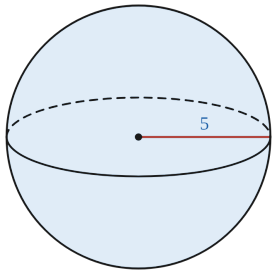
Hemisphere — half of a sphere, formed when a great circle _____ a sphere into two congruent halves

Part 1 · Notes and examples (fill in each note, then work the example)

Surface area of a sphere: S = _____

Example 1

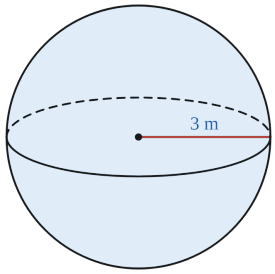
Find the surface area of the sphere. Round to the nearest hundredth.



Volume of a sphere: V = _____

Example 2

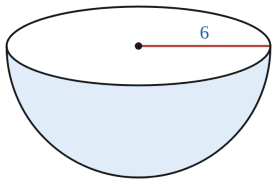
Find the volume of a sphere with a radius of 3 meters. Round to the nearest hundredth.



Hemisphere: the volume is half the sphere's; the surface area is half of $4\pi r^2$ _____ the flat circle, πr^2.

Example 3

Find the volume and the surface area of the hemisphere. Round to the nearest hundredth.



Part 2 · Partner practice

Partner practice

Find the surface area of a sphere with a radius of 7. Round to the nearest hundredth.